Introduction to Character Data-Type and String Data-Type

1. Character Data Type

The `char` data type in Java is used to store a single character. It is a 16-bit Unicode character that can hold any character from the Unicode character set.

Declaration and Initialization

```java
char letter = 'A';
char digit = '5';
char symbol = '@';
char space = ' ';
```

Key Points:

- Character literals are enclosed in single quotes (`'`)
- Each `char` occupies 2 bytes in memory
- Can store letters, digits, symbols, and special characters
- Unicode values range from 0 to 65,535

Examples:

```java
char grade = 'A';
char initial = 'J';
char newline = '\n';  // Escape sequence for newline
char tab = '\t';      // Escape sequence for tab
```

2. Manipulating Character Data-Type

Java provides several ways to work with character data, including converting between characters and their numeric representations.

Converting Characters to ASCII Values:

```java
char ch = 'A';
int asciiValue = (int) ch;
System.out.println("ASCII value of " + ch + " is: " + asciiValue);
// Output: ASCII value of A is: 65
```

Converting ASCII Values to Characters:

```java
int ascii = 97;
char character = (char) ascii;
System.out.println("Character for ASCII " + ascii + " is: " + character);
// Output: Character for ASCII 97 is: a
```

Character Comparison:

```java
char ch1 = 'A';
char ch2 = 'B';
if (ch1 < ch2) {
    System.out.println(ch1 + " comes before " + ch2);
}
```

3. Character.getNumericValue()

The `Character.getNumericValue()` method returns the numeric value of a character. This is particularly useful when working with digit characters.

Syntax:

```java
int numericValue = Character.getNumericValue(char ch);
```

Examples:

```java
char digitChar = '7';
int number = Character.getNumericValue(digitChar);
System.out.println("Numeric value: " + number);
// Output: Numeric value: 7

char letterChar = 'A';
int value = Character.getNumericValue(letterChar);
System.out.println("Numeric value of A: " + value);
// Output: Numeric value of A: 10
```

Practical Use:

```java
char[] digits = {'1', '2', '3', '4', '5'};
int sum = 0;
for (char digit : digits) {
    sum += Character.getNumericValue(digit);
}
System.out.println("Sum: " + sum);
// Output: Sum: 15
```

4. charAt()

The `charAt()` method is used to access a specific character at a given index in a String. Index starts from 0.

Syntax:

```java
char character = stringName.charAt(index);
```

Examples:

```java
String word = "Hello";
char firstChar = word.charAt(0);    // 'H'
char thirdChar = word.charAt(2);    // 'l'
char lastChar = word.charAt(4);     // 'o'

System.out.println("First character: " + firstChar);
System.out.println("Third character: " + thirdChar);
System.out.println("Last character: " + lastChar);
```

Iterating Through a String:

```java
String text = "Java";
for (int i = 0; i < text.length(); i++) {
    char ch = text.charAt(i);
    System.out.println("Character at index " + i + ": " + ch);
}
```

Error Handling:

```java
String str = "Programming";
// This would cause StringIndexOutOfBoundsException
// char ch = str.charAt(15); // Index out of bounds
```

5. Taking in String Data Type as Input

To read String input from the user, we use the `Scanner` class from the `java.util` package.

Import Statement:

```java
import java.util.Scanner;
```

Reading Complete Lines:

```java
Scanner scanner = new Scanner(System.in);
System.out.print("Enter your full name: ");
String fullName = scanner.nextLine();
System.out.println("Hello, " + fullName);
```

Reading Single Words:

```java
Scanner scanner = new Scanner(System.in);
System.out.print("Enter a word: ");
String word = scanner.next();
System.out.println("You entered: " + word);
```

Complete Example:

```java
import java.util.Scanner;

public class StringInput {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your favorite color: ");
        String color = scanner.nextLine();

        System.out.print("Enter your age: ");
        int age = scanner.nextInt();
        scanner.nextLine(); // Consume the newline

        System.out.print("Enter your city: ");
        String city = scanner.nextLine();

        System.out.println("Color: " + color);
        System.out.println("Age: " + age);
        System.out.println("City: " + city);
    }
}
```

6. str.length()

The `length()` method returns the number of characters in a String, including spaces and special characters.

Syntax:

```java
int length = stringName.length();
```

Examples:

```java
String message = "Hello World";
int length = message.length();
System.out.println("Length of the string: " + length);
// Output: Length of the string: 11

String empty = "";
System.out.println("Length of empty string: " + empty.length());
// Output: Length of empty string: 0

String withSpaces = "Java Programming";
System.out.println("Length: " + withSpaces.length());
// Output: Length: 16
```

Practical Applications:

```java
String password = "myPassword123";
if (password.length() >= 8) {
    System.out.println("Password length is acceptable");
} else {
    System.out.println("Password is too short");
}
```

Using with charAt():

```java
String text = "Programming";
int lastIndex = text.length() - 1;
char lastCharacter = text.charAt(lastIndex);
System.out.println("Last character: " + lastCharacter);
// Output: Last character: g
```

Quiz Questions

Question 1: What is the correct way to declare and initialize a character variable in Java? a) `char ch = "A";` b) `char ch = 'A';` c) `char ch = A;` d) `character ch = 'A';`

Answer: b) `char ch = 'A';`

Question 2: What will `Character.getNumericValue('5')` return? a) '5' b) 5 c) 53 d) Error

Answer: b) 5

Question 3: If a String `str = "Java"`, what will `str.charAt(2)` return? a) 'J' b) 'a' c) 'v' d) 'a'

Answer: c) 'v'

Question 4: What is the length of the String `"Hello World"`? a) 10 b) 11 c) 12 d) 9

Answer: b) 11

Question 5: Which Scanner method should be used to read a complete line of text including spaces? a) `next()` b) `nextLine()` c) `nextString()` d) `readLine()`

Answer: b) `nextLine()`

Practice Coding Questions

Question 1: Write a Java program that takes a character as input from the user and displays its ASCII value.

Question 2: Create a program that reads a string from the user and prints each character along with its index position.

Question 3: Write a program that takes a string as input and counts how many digits (0-9) are present in the string using `Character.getNumericValue()`.

Question 4: Develop a program that reads a person's full name and displays the first and last characters of the name.

Question 5: Create a program that takes a string input and checks if it's a palindrome by comparing characters from both ends using `charAt()` method.